

Eric Ding

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EDUCATION

Cornell University

Ph.D., Computer Science, Direction: Systems and Networking, GPA: 4.0

Ithaca, NY, United States

Aug. 2024 – June 2029 Expected

University of Michigan

B.S.E., Computer Science, Summa Cum Laude, GPA: 3.96, Dual Degree Program

Ann Arbor, MI, United States

Aug. 2022 – Apr. 2024

Shanghai Jiao Tong University

B.S.E., Electrical and Computer Engineering, Outstanding Graduate, GPA: 3.76, top 10%

Shanghai, China

Sep. 2020 – Aug. 2024

RESEARCH EXPERIENCE

Graduate Student Researcher

Cornell University

Ithaca, New York

Aug. 2024 – Present

- Designed RDMA networking protocols and ML workload simulator for energy-efficient GPU interconnects.
- Built profiling and benchmarking pipelines for evaluating collective communication implementation.
- Implemented straggler-mitigation collective communication algorithm for distributed ML workloads.
- Developed physical layer routing algorithms for silicon photonic fabrics in rack-scale ML networks.

Machine Learning System Research Assistant

SymbioticLab, University of Michigan

Ann Arbor, MI, United States

May 2023 – Apr. 2024

- Developed resource management systems for federated learning.
- Contributed to open-source project *FedScale* in federated learning optimizer implementation.

Embedded System Research Assistant

The Fan Lab, University of Michigan

Ann Arbor, MI, United States

May 2023 – Sep. 2023

- Designed a reliable communication protocol that operates atop I2C and Bluetooth Low Energy protocols.

WORK EXPERIENCE

Research Intern, Networking

NVIDIA

Santa Clara CA, US

May 2026 – Aug. 2026

- Designed a network architecture for GPU NVLink scale-up domain for LLM inference workload acceleration.

Artificial Intelligence Internship

Bosch

Shanghai, China

May 2024 – Aug. 2024

- Designed and implemented a multi-agent system using LangGraph, OpenAI endpoints, and Neo4J graph database for retrieval augmented generation and domain-specific task reasoning.

PUBLICATION

- **Eric Ding**, Barry Lyu, Bhaskar Kataria, Rachee Singh, Opus: Photonic Rail-Optimized Fabric in ML Datacenters, ACM Special Interest Group on Data Communications (**SIGCOMM 2026**)
- Abhishek Vijaya Kumar, **Eric Ding**, Arjun Devraj, Darius Bunandar, Rachee Singh, Reconfigurable Torus Fabrics for Multi-tenant ML, ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS, 2026**)
- **Eric Ding**, Chuhan Ouyang, Rachee Singh, Photonic Rails in ML Datacenters, Twenty-Fourth ACM Workshop on Hot Topics in Networks (**HotNets 2025**)
- **Eric Ding**, Rachee Singh, PipSwitch: A Circuit Switch Using Programmable Integrated Photonics, Optical Fiber Communication Conference (**OFC 2025**)
- Jiachen Liu, Fan Lai, **Eric Ding**, Yiwen Zhang, Mosharaf Chowdhury, Venn: Resource Management Across Federated Learning Jobs, The Eighth Annual Conference on Machine Learning and Systems (**MLSys 2025**)
- Anjali Devi Sivakumar, Ruchi Sharma, Chandrakalavathi Thota, **Eric Ding**, Xudong Fan, WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss, **ACS Sensors 2023**

- **Eric Ding**, Byungsoo Oh, Bhaskar Kataria, Kaiwen Guo, Jelena Gvero, Abhishek Vijaya Kumar, Arjun Devraj, Lindsey Bowen, Atharv Sonwane, Emaad Manzoor, Rachee Singh, CCL-Bench 1.0: A Trace-Based Benchmark for LLM Infrastructure, Under submission
- Arjun Devraj, **Eric Ding**, Abhishek Vijaya Kumar, Robert Kleinberg, Rachee Singh, Efficient AllReduce with Stragglers, Under submission

VOLUNTEERING

- NSDI 2026 Fall AE Committee, *Jan. 2026*
- EuroSys 2026 Shadow PC, *Oct. 2025*
- SIGCOMM 2025 AE Committee, *Aug. 2025*
- MLSys 2025 AE Reviewer, *Mar. 2025*
- Lunch and Learn Czar, *Computer Science Department, Cornell University, Feb. 2025 - May. 2025*
- Computer Science Peer Tutor, *School of Information, University of Michigan, Feb. 2024 - May. 2024*

TEACHING EXPERIENCE

- TA, Systems for Large-scale ML (CS5470), Cornell University, Fall 2025. Responsible for assignment development: AI inference performance benchmarking, request scheduling and batching mechanisms, CUDA kernel optimization.

PROJECTS

A Fully Synthesizable Out-of-Order N-Way RISC-V Processor | *SystemVerilog, Verdi*

CNN Convolution Optimization | *CUDA, Slurm*

Operating System Kernel (CPU Scheduler, Thread Library, Memory Management, File System) | *C++*

Distributed Search Engine (MapReduce, PageRank) | *Python, React, AWS*

GRANTS AND AWARDS

- Graduate School Fellowship, Cornell University, Feb. 2024
- James B. Angell Scholar, University of Michigan, Feb. 2024
- University Honors, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Dean's List, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Tang Junyuan JI Scholarship Nominee, Aug. 2022
- Shanghai Jiao Tong University Merit Student Award, Nov. 2021

TECHNICAL SKILLS

Languages: C++, Python, C, SystemVerilog, Go, RISC-V, Matlab, Javascript, Bash, SQL, R

Frameworks: CUDA, NCCL, PyTorch, Tensorflow, JAX, vLLM, DeepSpeed, TorchTitan, LangChain, React, gRPC, Hadoop

Developer Tools: Git, Docker, Kubernetes, Redis, Nvidia Nsight, L^AT_EX, Neo4J, VSCode, Arduino, STM32CubeIDE, Wireshark, Gurobi

Simulation and Modeling: Verdi, Catia, Matlab, Mathematica, LabVIEW, Pspice, Proteus, Vivado, Astra-Sim, ns-3, gem5

Selected courses: EECS470 Computer Architecture, EECS 471 Applied Parallel Programming with GPUs, EECS482 Operating System, EECS489 Computer Network